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The continuous-coefficient Jensen equation: A note on three vestigial regularity hypotheses

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The continuous-coefficient Jensen equation: A note on three vestigial regularity hypotheses

Double Blind No Name

Abstract. Jensen’s inequality says that for a concave function G and a random variable ξ on an interval, $\mathbb{E}[G(\xi)] \leq G(\mathbb{E}[\xi])$. A natural question: if equality holds for every two-point distribution—if $pG(u_1) + (1-p)G(u_2) = G(pu_1 + (1-p)u_2)$ for all u_1, u_2 in the interval and all weights $p \in [0, 1]$ —must G be affine?

The answer is yes. The proof is one line: substitute $u_1 = M$, $u_2 = 0$, $p = v/M$ to read $G(v)$ directly off the chord through the endpoints. No regularity hypothesis on G —continuity, measurability, monotonicity, or boundedness—is needed.

This contrasts with a celebrated 80-year story. The $p = \frac{1}{2}$ specialization is equivalent to Cauchy’s additive equation and admits wildly pathological non-affine solutions without regularity. The Polish school of the 1920s cataloged which tameness conditions—continuity, measurability, boundedness on a set of positive measure—suffice to rule out the pathology. These conditions became standard cargo in functional-equation arguments, carried reflexively even where the full continuous-weight equation is in play and no such cargo is needed.

We make explicit which classical hypotheses become vestigial once all weights $p \in [0, 1]$ are tested (rather than rational weights only), explain structurally why (the pathology lives at irrational p , precisely where the continuous equation tests G), and identify the recurring trap: derivations in expected-utility theory, entropy axiomatics, and surrogate calibration that saturate Jensen across the full two-point family each produce this equation and each benefit from the one-line closure.

Keywords: Jensen equation, Cauchy equation, Hamel basis, functional equation, regularity hypothesis

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1. INTRODUCTION. One of the most serviceable inequalities in mathematics—Jensen’s inequality, $\mathbb{E}[G(\xi)] \leq G(\mathbb{E}[\xi])$ for concave G —raises an immediate follow-up: when does equality hold for every two-point distribution supported on a given interval? The equation expressing universal two-point equality,

$$pG(u_1) + (1-p)G(u_2) = G(pu_1 + (1-p)u_2) \quad (u_1, u_2 \in I, p \in [0, 1]), \quad (\star)$$

imposed on a function $G: I \rightarrow \mathbb{R}$ on a real interval I , has at least three faces in classical analysis. Those faces have been steadily conflated for over a century, with consequences for both the foundational theory and the modern applied literature where the equation surfaces.

Three faces of the equation. The *discrete-coefficient* Jensen equation,

$$G\left(\frac{u_1+u_2}{2}\right) = \frac{G(u_1)+G(u_2)}{2}, \quad (J_2)$$

is the $p = \frac{1}{2}$ specialization of $(\star)$. Setting $f(x) := G(x) - G(0)$, the equation (J_2) is equivalent on a translate of I to Cauchy’s additive equation $f(x+y) = f(x) +$

$f(y)$. Cauchy's equation inherits a full apparatus of pathological solutions: without measurability, monotonicity, boundedness on a set of positive Lebesgue measure, or another regularity hypothesis, the equation admits non-affine solutions constructed via a *Hamel basis*—a basis of $\mathbb{R}$ as a $\mathbb{Q}$ -vector space, whose existence requires the axiom of choice. The classical pathological-solution apparatus is the work of Hamel [1]; the regularity hypotheses sufficient to recover affineness were cataloged by Cauchy [2], Darboux [3], Sierpiński [4], Steinhaus [5], and Ostrowski [6].

The *rational-coefficient* Jensen equation,

$$pG(u_1) + (1-p)G(u_2) = G(pu_1 + (1-p)u_2) \\ (u_1, u_2 \in I, \quad p \in [0, 1] \cap \mathbb{Q}), \quad (J_{\mathbb{Q}})$$

strengthens (J_2) but does not escape its pathology. Cauchy additivity entails $\mathbb{Q}$ -homogeneity— $f(qx) = qf(x)$ for every $q \in \mathbb{Q}$ and $x \in \mathbb{R}$ —by an elementary induction on positive integers, sign reversal for negatives, and the inverse-of-multiplication argument for $1/n$. Combining additivity with $\mathbb{Q}$ -homogeneity returns $(J_{\mathbb{Q}})$ for f , hence for G . So $(J_{\mathbb{Q}})$, (J_2) , and Cauchy's equation share the same solution class up to constants, and all three inherit the Hamel-basis pathology.

The *continuous-coefficient* form $(\star)$, by contrast, is fundamentally different. The strengthening from $\mathbb{Q}$ to $\mathbb{R}$ in the range of p is *genuine*. As we will see, $(\star)$ does not admit any Hamel pathology: the pathology cannot survive at irrational p , and the continuous-coefficient equation tests G at every irrational. This is the subject of the paper.

The main result.

Theorem 1. *Let $M > 0$ and let $G: [0, M] \rightarrow \mathbb{R}$ satisfy $(\star)$ for every $u_1, u_2 \in [0, M]$ and every $p \in [0, 1]$. Then G is affine on $[0, M]$:*

$$G(v) = \frac{G(M) - G(0)}{M} v + G(0), \quad v \in [0, M].$$

No regularity hypothesis on G (continuity, measurability, monotonicity, boundedness) is required.

The proof is one line: a substitution into $(\star)$ that pins $G(v)$ along the chord through the endpoints $(0, G(0))$ and $(M, G(M))$. We give it in Section 2 and use the remainder of the paper to articulate what it means.

The historical detour and the structural moral. Theorem 1 is folklore (cf. [7, §2.1], [8, Ch. 13]). It is *not* a new mathematical result. The contribution of the present paper is in three parts:

- *The explicit dictionary* (Section 3) of three regularity hypotheses on G —continuity, measurability, boundedness—that are required to force affineness for (J_2) but become *vestigial* under $(\star)$, together with the structural mechanism (the Hamel pathology lives at irrational p , exactly where $(\star)$'s continuous coefficient closes the door).
- *The strict-minimum hypothesis* (Section 4)—the equation $(\star)$ at a single chord configuration already suffices.
- *The recurrence pattern* (Section 5)—the trap of invoking unnecessary Cauchy–Hamel regularity recurs whenever a derivation pushes Jensen's inequality to saturation across a wide class of two-point distributions. We survey three example application areas where this pattern arises.

The 80-year detour through Hamel and the Polish school answered a perfectly natural question—*what regularity hypothesis suffices for $(J_2) \Rightarrow \text{affine}$?*—but a *different* question from the one that matters for many modern applications. The applied derivation does not produce (J_2) or $(J_{\mathbb{Q}})$; it produces $(\star)$ as a saturated identity, and the one-line substitution of Section 2 is the right closure tool. The dictionary of Section 3 makes explicit which classical regularity hypotheses become vestigial in this transition.

Terminology. Throughout the paper we refer to the substitution $u_1 = M, u_2 = 0, p = v/M$ in $(\star)$ —by which G is pinned along the chord through the endpoints $(0, G(0))$ and $(M, G(M))$ —as the *chord substitution*. The substitution is standard folklore in the functional-equations literature (cf. [7, §2.1], [8, Ch. 13]); the descriptive label is our convenience, adopted for brevity in repeated reference rather than as a claim of established terminology.

2. THE RESULT AND ITS PROOF.

Proof of Theorem 1. Fix $v \in [0, M]$. Set

$$u_1 := M, \quad u_2 := 0, \quad p := \frac{v}{M}.$$

The constraints of $(\star)$ are satisfied: $u_1 = M \in [0, M]$, $u_2 = 0 \in [0, M]$, and $p = v/M \in [0, 1]$ since $v \in [0, M]$. Substituting into $(\star)$,

$$\frac{v}{M} G(M) + \left(1 - \frac{v}{M}\right) G(0) = G\left(\frac{v}{M} \cdot M + \left(1 - \frac{v}{M}\right) \cdot 0\right) = G(v).$$

Rearranging yields $G(v) = G(0) + (G(M) - G(0))v/M$. ■

The chord substitution pins $G(v)$ along the straight chord from $(0, G(0))$ to $(M, G(M))$ for every $v \in [0, M]$. Since this determines $G(v)$ at every v , the function G is fully determined by its values at the two endpoints—it is exactly the affine function joining them.

Corollary 2. *Under the hypotheses of Theorem 1, G is in particular continuous, monotone, locally Lipschitz, absolutely continuous, and Lebesgue measurable on $[0, M]$.*

Proof. Every affine function $G(v) = av + b$ on a bounded interval is differentiable with constant derivative $G' = a$, hence continuous; non-decreasing if $a \geq 0$ and non-increasing if $a \leq 0$; Lipschitz with constant $|a|$; absolutely continuous; and Borel-measurable, hence Lebesgue-measurable. ■

The order of inference matters and is worth stating explicitly. Under $(\star)$, the regularity properties of G are the *conclusion*, not the hypothesis. Each regularity hypothesis one might be tempted to assume—continuity, monotonicity, boundedness, measurability—is automatically satisfied by any solution of $(\star)$. The continuous-coefficient equation supplies its own regularity from within; no external regularity hypothesis is needed.

This is what makes the continuous-coefficient case structurally different from the discrete-coefficient case (J_2) , where some regularity hypothesis *is* required to rule out the Hamel pathology. We now examine the difference in detail.

3. THE DICTIONARY OF VESTIGIAL REGULARITY HYPOTHESES.

Six hypotheses and their fates. We place side-by-side the discrete-coefficient equation (J_2) and the continuous-coefficient equation $(\star)$, and ask which regularity hypotheses on G are *required* to force affineness for each. The first column of Table 1 names the hypothesis; the second records the requirement for (J_2) with the standard historical attribution; the third records what becomes of the requirement under $(\star)$.

Hypothesis on G	For $(J_2) \Rightarrow$ affine?	For $(\star) \Rightarrow$ affine?
Continuity on I: G is continuous at every $v \in I$.	Yes, suffices (Cauchy [2]).	No —it is the <i>conclusion</i> (Corollary 2).
Measurability on I: G is Lebesgue- or Borel-measurable on I .	Yes, suffices (Sierpiński [4]).	No —same.
Monotonicity on I: G is non-decreasing or non-increasing on I .	Yes, suffices (Darboux [3]).	No —same.
Boundedness on a set of positive measure: G is bounded on some Lebesgue-measurable $E \subseteq I$ with $ E > 0$.	Yes, suffices (Steinhaus [5]).	No —same.
Boundedness on I: G is bounded on all of I .	Yes, suffices (Ostrowski [6]).	No —same.
None: no regularity hypothesis at all.	Insufficient —Hamel-basis pathology [1].	Sufficient —Theorem 1.

Table 1. Regularity hypotheses on G . Column 2 records which hypotheses are classically required to force affineness for the discrete-coefficient equation (J_2) . Column 3 records their status under the continuous-coefficient equation $(\star)$.

The bottom row is the point of the table. **No regularity hypothesis** suffices for (J_2) : there exists a non-affine, non-measurable, unbounded, additive function on $\mathbb{R}$ —the classical Hamel-basis pathology [1]—that supplies a non-affine solution of (J_2) . Conversely, **no regularity hypothesis** is *needed* for $(\star)$: Theorem 1’s chord substitution closes the proof at the level of the bare functional equation.

The five non-bottom rows tell a complementary story. For (J_2) , *some* hypothesis from these five is required; otherwise the Hamel pathology survives. For $(\star)$, *none* of these is required—and yet each is automatically satisfied by any solution (Corollary 2). Each hypothesis is, in the dictionary’s terminology, *vestigial*: classically required for (J_2) , classically expected for $(\star)$, but in fact unnecessary for $(\star)$.

The structural mechanism. Why does the Hamel pathology fail to apply to $(\star)$? The mechanism is direct and illuminating.

A Hamel-basis pathological solution G of Cauchy’s equation on $\mathbb{R}$ is $\mathbb{Q}$ -linear: $G(0) = 0$ and $G(qx) = qG(x)$ for every $q \in \mathbb{Q}$ and $x \in \mathbb{R}$. By design, G is *not* $\mathbb{R}$ -linear: there exists an irrational $r \in \mathbb{R}$ and an $x \in \mathbb{R}$ such that $G(rx) \neq rG(x)$.

The chord identity in $(\star)$ at the configuration $u_1 = M, u_2 = 0$ reads

$$pG(M) + (1-p)G(0) = G(pM). \quad (\star_0)$$

For a Hamel-pathological G (so $G(0) = 0$), $(\star_0)$ becomes $pG(M) = G(pM)$ —exactly the $\mathbb{R}$ -linearity assertion for G at the pair (M, p) . By $\mathbb{Q}$ -linearity of G , this holds for every $p \in \mathbb{Q}$. By the failure of $\mathbb{R}$ -linearity, it *fails* at some irrational p . So the Hamel-pathological G satisfies $(J_{\mathbb{Q}})$ (every rational coefficient is OK) but violates $(\star)$ at every irrational p where the $\mathbb{R}$ -linearity defect appears.

The structural point is one sentence:

The Hamel pathology lives at irrational p ; the continuous-coefficient equation $(\star)$ fore-closes on it there.

The 80-year detour through Hamel and the Polish school—Cauchy 1821 to Ostrowski 1929—was an investigation of what regularity hypothesis can substitute for the *unavailable* constraint at irrational p , when only the rational-coefficient version of the equation is in play. With the continuous-coefficient version, the irrational-coefficient constraint is *available*, and the entire dictionary collapses.

4. VARIANTS AND LIMITS.

The strict-minimum hypothesis. The proof of Theorem 1 uses $(\star)$ at exactly one configuration: $u_1 = M, u_2 = 0$, with $p \in [0, 1]$ ranging freely. The full force of $(\star)$ —for every pair (u_1, u_2) —is not consumed. We record the strict minimum.

Theorem 3. *Let $M > 0$ and let $G: [0, M] \rightarrow \mathbb{R}$ satisfy*

$$pG(M) + (1-p)G(0) = G(pM) \quad \text{for all } p \in [0, 1].$$

Then $G(v) = G(0) + (G(M) - G(0))v/M$ on $[0, M]$.

Proof. Set $p := v/M$; then $p \in [0, 1]$ since $v \in [0, M]$. The hypothesis gives

$$pG(M) + (1-p)G(0) = G(pM) = G(v),$$

so $G(v) = G(0) + p(G(M) - G(0)) = G(0) + (G(M) - G(0))v/M$. ■

Theorem 3 is *in principle* weaker than Theorem 1: the one-chord hypothesis does not a priori imply the full $(\star)$. Under the conclusion (affineness) both hold; in practice an author who derives $(\star)$ from a richer setup (e.g., a Jensen-equality identity in a Bayes-risk computation; see Section 5) typically obtains $(\star)$ in full force. The narrower Theorem 3 is the right reference when one wants to know exactly how much of the equation the proof actually consumes.

Convex domains in higher dimensions. The chord substitution extends to any convex subset of a real vector space.

Theorem 4. *Let V be a real vector space, let $C \subseteq V$ be a convex set, and let $G: C \rightarrow \mathbb{R}$ satisfy*

$$pG(x_1) + (1-p)G(x_2) = G(px_1 + (1-p)x_2) \quad (x_1, x_2 \in C, p \in [0, 1]).$$

Then there is a linear functional $a: \text{span}(C - C) \rightarrow \mathbb{R}$ and a constant $b \in \mathbb{R}$ such that $G(x) = a(x - x_0) + b$ on C for any fixed $x_0 \in C$. Equivalently, G is affine on C .

Proof sketch. Apply Theorem 1 along every chord in C : for each direction v with $x_0 + v \in C$, the map $\lambda \mapsto G(x_0 + \lambda v)$ is affine in $\lambda \in [0, 1]$. Hence $G(x_0 + v) - G(x_0)$ depends linearly on v in each direction; an inductive argument on the dimension of the affine span of the points involved promotes this to a global linear functional a on $\text{span}(C - C)$. The bookkeeping is standard and is given in [9, Ch. 13]. ■

The structurally essential step is the one-dimensional chord substitution of Theorem 1. The higher-dimensional lifting is bookkeeping.

The rational-coefficient version retains the pathology. The strengthening from rational to continuous coefficients in $(\star)$ is doing genuine work: the rational-coefficient version $(J_{\mathbb{Q}})$ retains the Hamel pathology.

Proposition 5 (Folklore; cf. [8, Ch. 5]). *There exists $G: [0, 1] \rightarrow \mathbb{R}$ satisfying $(J_{\mathbb{Q}})$ (hence (J_2)) on $[0, 1]$ that is not affine.*

Construction. Choose a Hamel basis H of $\mathbb{R}$ over $\mathbb{Q}$ containing 1; such a basis exists by Zorn's lemma applied to the partially ordered set of $\mathbb{Q}$ -linearly independent subsets of $\mathbb{R}$. Since H has cardinality $\mathfrak{c}$ and $\mathbb{Q} \cup \{1\}$ is countable, H contains an irrational $h \in (0, 1)$. Define a $\mathbb{Q}$ -linear map $\ell: \mathbb{R} \rightarrow \mathbb{R}$ by setting $\ell(1) = 0$, $\ell(h) = 1$, and extending $\mathbb{Q}$ -linearly on the remaining basis elements.

By construction, ℓ is $\mathbb{Q}$ -linear: $\ell(x + y) = \ell(x) + \ell(y)$ (Cauchy's equation) and $\ell(qx) = q\ell(x)$ for all $q \in \mathbb{Q}$ ($\mathbb{Q}$ -homogeneity). Set $G := \ell|_{[0,1]}$; then G satisfies $(J_{\mathbb{Q}})$ on $[0, 1]$ because $\mathbb{Q}$ -rational convex combinations of points of $[0, 1]$ remain in $[0, 1]$ and ℓ respects $\mathbb{Q}$ -linear combinations.

Explicit witness that G is not affine: $G(h) = \ell(h) = 1 \neq 0$, while $G(q) = \ell(q) = q\ell(1) = 0$ for every rational $q \in [0, 1]$. Any affine map $A: [0, 1] \rightarrow \mathbb{R}$ agreeing with G on $\mathbb{Q} \cap [0, 1]$ would satisfy $A(0) = 0$ and $A(q) = 0$ for all rational q —forcing $A \equiv 0$, contradicting $G(h) = 1$. ■

Proposition 5 sharpens the dictionary: $(J_{\mathbb{Q}})$ on $[0, 1]$ admits non-affine solutions; only the strengthening to the continuous-coefficient $(\star)$ retires the regularity-hypothesis requirement.

5. WHERE THE TRAP RECURS: JENSEN SATURATION AS A STRUCTURAL PATTERN. The chord substitution of Section 2 is folkloric—both [7, §2.1] and [8, Ch. 13] articulate the underlying observation, and it is standard in the functional-equations community. Why, then, does this paper exist?

Because the explicit packaging matters. The functional-equations community has internalized the chord substitution; the applied communities that produce the continuous-coefficient equation as a byproduct of Jensen saturation have not. These vestigial regularity hypotheses—continuity, measurability, and boundedness—keep reappearing in applied derivations of $(\star)$ outside the functional-equations community. The trap is not a one-off oversight; it has a structural source. This section identifies the source and surveys three application areas where it surfaces.

The structural source: saturated Jensen. The recurrence is predictable. Whenever a derivation in any field arrives at an identity of the form

$$\mathbb{E}[g(\xi)] = g(\mathbb{E}[\xi]) \quad (\xi \text{ a random variable on } I, g: I \rightarrow \mathbb{R}), \quad (\dagger)$$

for a class of random variables ξ wide enough that the marginal $\mathbb{E}[\xi]$ can be any point of I and the support of ξ can be any two-point subset of I with any pair of

masses $(p, 1 - p)$, $p \in [0, 1]$, the identity is exactly $(\star)$ with $g = G$ and ξ supported on $\{u_1, u_2\}$ with masses $\{p, 1 - p\}$. Such an identity arises whenever **Jensen's inequality is pushed to saturation**—equality is required in $\mathbb{E}[g(\xi)] \leq g(\mathbb{E}[\xi])$ for concave g across the full two-point family.

Jensen's inequality is the central tool in dozens of fields. In *most* of these fields, Jensen is used as an inequality with explicit slack (tracked in convex analysis via biconjugates and Fenchel duality; in calibration theory via the ψ -transform of Bartlett, Jordan, and McAuliffe [10]; in entropy via convexity arguments). Such derivations never produce $(\star)$ as a saturated identity.

In *some* fields, a derivation produces $(\dagger)$ as a saturated identity: the slack is forced to zero by a structural property of the setup (e.g., the class of ξ is rich enough to exhaust the inequality's freedom). At that moment $(\star)$ appears, and the chord substitution is the natural closure. An author unfamiliar with the chord substitution will reach for the classical Cauchy–Hamel toolkit and find that the regularity hypothesis is unnecessary only after the fact.

Application: expected-utility representation theorems. In **expected-utility theory** in the von Neumann–Morgenstern tradition [11], the 1944 axiomatization of preferences over lotteries—sharpened by Herstein and Milnor [12]—derives a utility functional U on a space of lotteries $\mathcal{L}$ that is *linear in probability*:

$$U(pL_1 + (1 - p)L_2) = pU(L_1) + (1 - p)U(L_2) \quad (L_1, L_2 \in \mathcal{L}, p \in [0, 1]).$$

This is exactly $(\star)$ with U as G and lotteries as points. The classical closure of the linearity step uses the *Archimedean axiom*: for any three lotteries $L_1 \succ L_2 \succ L_3$, there exist $p, q \in (0, 1)$ such that $pL_1 + (1 - p)L_3 \succ L_2 \succ qL_1 + (1 - q)L_3$. The Archimedean axiom forces continuity-in-probability of preferences, which in turn forces continuity of U .

But the chord substitution of Theorem 1 *also* closes the linearity step, without any continuity hypothesis. The chord substitution is the algebraic alternative to the Archimedean axiom. Even where the Archimedean axiom is invoked (and rightly so, as it carries other preference-theoretic content), the closure of the linearity equation $(\star)$ itself does not require it.

Application: Shannon entropy's axiomatic characterization. In the **axiomatic characterization of Shannon entropy** in the tradition of Khinchin [13] and Faddeev [14], Shannon's entropy function $H(p_1, \dots, p_n) = -\sum_i p_i \log p_i$ is characterized by three axioms: continuity, maximum at the uniform distribution, and a *recursivity axiom* that relates the entropy of a joint distribution to a marginal entropy plus a conditional.

In the modern formalization [7, Ch. 5], the recursivity axiom reduces, in intermediate steps, to a system of functional equations on auxiliary functions g, h on $[0, 1]$. Several of these intermediate equations are of $(\star)$ -form for the auxiliary functions, and the closure step uses continuity (the first Khinchin axiom). The chord substitution of Theorem 1 is the alternative closure that does not require continuity as a hypothesis; in Aczél–Dhombres [9] the substitution is part of the standard toolkit.

Application: surrogate calibration on the resolution axis. In **surrogate calibration** in statistical decision theory, the foundational tradition—Bartlett, Jordan, and McAuliffe [10] and its extensions by Tewari and Bartlett [15], Steinwart [16], and Reid and Williamson [17, 18]—derives its results via convex analysis (biconjugates, ψ -transforms, Fenchel duality) and consequently tracks Jensen as an *inequality* with explicit slack. These derivations never produce $(\star)$ as a saturated identity.

In a recent manuscript of the author [19] on the achievable error floor of partition-based classifiers, the resolution-axis transposition framing of surrogate calibration produces $(\star)$ directly: a two-cell partition computation forces $(\star)$ on the function G expressing the partition Bayes risk in terms of an aggregated concave score, with the cell mass p ranging freely over $[0, 1]$ on an atomless probability space. In the Lean 4 formalization of [19], the corresponding lemma was initially declared with a boundedness hypothesis on G in deference to the Cauchy–Hamel literature; the proof body then found that hypothesis unused, by the chord substitution of Theorem 1. This episode is described in the Appendix.

The lesson generalizes. Any derivation that saturates Jensen across a wide class of two-point distributions should be expected to produce $(\star)$ and to benefit from Theorem 1’s one-line closure.

An invitation. The three application areas above are not exhaustive. We invite readers who have encountered $(\star)$ in their own work—in mathematical economics, information theory, statistical decision theory, physics, or elsewhere—and who have invoked an unnecessary Cauchy–Hamel regularity hypothesis to extend the catalog. The structural pattern of saturated Jensen and the chord substitution as its natural closure is, we believe, recurrent enough to merit explicit articulation.

6. CONCLUDING REMARKS. Theorem 1 is, mathematically, one line. The dictionary of Section 3 is six rows of a table. The structural argument of Section 3 is one sentence: *the Hamel pathology lives at irrational p ; the continuous-coefficient equation $(\star)$ forecloses on it there.* Why does this material deserve an article in the MONTHLY?

Two reasons. First, the 80-year detour through Hamel and the Polish school is a *foundational* result of classical analysis—and the dictionary of Section 3 articulates a precise sense in which the detour answered an adjacent question: not “what is the minimum hypothesis for $(\star) \Rightarrow \text{affine}$?” (Theorem 1’s answer: nothing), but “what is the minimum hypothesis for $(J_2) \Rightarrow \text{affine}$?” (Polish school’s answer: any of a dictionary of tameness conditions). Both questions are interesting; their answers are different.

Second, the recurrence pattern of Section 5 is, to our knowledge, not articulated explicitly elsewhere in the functional-equations literature. The pattern—saturated Jensen produces $(\star)$, and the chord substitution closes it—is recognizable in a number of modern application contexts (utility theory, entropy, calibration), and the explicit articulation may save other authors from carrying vestigial regularity hypotheses in their own work.

The chord substitution itself is the kind of one-line proof that, once seen, is hard to unsee—but that nonetheless takes some scaffolding to make visible. The present paper is the scaffolding.

A. HOW THE OBSERVATION CAME UP: A NOTE ON PROVENANCE. *This appendix is non-mathematical and may be skipped by readers interested only in the mathematics. It describes the route by which the observation surfaced—through a proof-assistant exercise—and may interest readers unfamiliar with how mechanized proof can pressure-test classical heuristics. Readers interested only in the mathematics can stop at Section 6.*

The starting position: a defensive hypothesis added on purpose. The chord substitution surfaced during the formalization of a separate paper by the present author [19] in the **Lean 4** proof assistant, paired with the **Mathlib** mathematical library. The relevant lemma expressed the step: *if $G: [0, M] \rightarrow \mathbb{R}$ satisfies the continuous-coefficient Jensen*

equation $(\star)$, then G is affine.

When the lemma was first stated in informal mathematics, the author added a **boundedness hypothesis** on G —the image $G([0, M])$ is a bounded subset of $\mathbb{R}$ —for textbook reasons: an internal adversarial-review round had correctly flagged that, for the *discrete-coefficient* equation (J_2) , no affineness conclusion holds without some regularity (Hamel’s pathology; cf. Section 3). Boundedness on I is a sufficient hypothesis by a classical theorem of Steinhaus and Ostrowski (Table 1, row 5). The author expected boundedness to be load-bearing and declared the Lean lemma accordingly, binding the boundedness witness to a named variable.

The discovery. The proof body, when written, did the only natural thing: it set $u_1 := M$, $u_2 := 0$, $p := v/M$. The equation rearranged in two lines to the explicit affine formula, and the proof closed—with the boundedness witness untouched. The proof assistant raised no error (an unused hypothesis is legal), but the author noticed.

Removing the boundedness witness from the lemma’s signature and re-checking: still valid. The hypothesis the author had carried defensively was vestigial.

The author’s first reaction was suspicion: the literature on Cauchy’s equation is unambiguous that *some* regularity is needed (Table 1). The working assumption had been that the same held here. The proof assistant said otherwise.

What the books say. The canonical references were then consulted: J. Aczél, *Lectures on Functional Equations and Their Applications*, §2.1 [7], and M. Kuczma, *An Introduction to the Theory of Functional Equations and Inequalities*, Ch. 13 [8]. Both books treat the continuous-coefficient Jensen equation $(\star)$ as a separable problem and observe—in compressed form, embedded in longer treatments of richer equations—that $(\star)$ admits the chord-substitution closure with no regularity hypothesis. The observation is not a discovery of the present paper: it has been part of the functional-equations toolkit since at least Aczél’s 1966 encyclopedia.

What is *not* standard is the **explicit packaging** of the observation as a separable lemma with an accompanying dictionary and a structural explanation of where the pathology lives. Aczél and Kuczma each devote no more than a paragraph to the observation; the dictionary of Section 3 and the recurrence pattern of Section 5 are, to the author’s knowledge, the contribution of the present paper.

The honest order of events is therefore: **proof assistant first, books second**. The author did not find the observation in Aczél or Kuczma and then formalize it; the author formalized the lemma defensively, the proof assistant showed the hypothesis was unused, and the literature then confirmed that the observation underlying the closure is known and properly attributed in the functional-equations community. This appendix is the record of that order.

Two modest morals. *A defensive hypothesis is not free.* In any application that needs the lemma, the user must verify the hypothesis. If the hypothesis is vestigial, that verification is wasted; worse, in a setting where the hypothesis is not naturally available, the user may abandon a correct proof route on the mistaken belief that a deeper closure tool is required. The dictionary of Section 3 is, in part, an inventory of regularity hypotheses that—under the continuous-coefficient form—are exactly such defensive additions.

Mechanized proof complements mathematical intuition. The proof assistant did not discover the chord substitution—Aczél and Kuczma had it long before—but it did force the author to notice that a hypothesis carried for textbook reasons was unused. That noticing prompted the literature search that produced the dictionary and the structural

pattern. Without the mechanization, the author would in all likelihood have written up the lemma with the boundedness hypothesis attached and never asked the question.

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During the preparation of this paper the author used an AI language model (Anthropic’s Claude Opus, accessed via GitHub Copilot) in two specific capacities: (i) adversarial review of successive drafts, in which the model was prompted in hostile-referee mode to surface defects in the prose, argument structure, and prior-art positioning; (ii) formatting assistance— $\text{\LaTeX}$ mechanics and bibliography construction. All mathematical content, including all theorems, proofs, and the structural analysis of Sections 3 and 5, was conceived, verified, and refined by the author, who takes full responsibility for the paper.

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